

What is Syzygies?

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We cover the following topics in this note.

- Syzygy
- TBA.

Linear Algebra	Commutative Algebra
Field $\mathbb{F}$	Commutative ring R
Vector space V over a field $\mathbb{F}$ ($\mathbb{F}$ -module)	Module M over a ring R (R -module)
Vectors $\mathbf{v}_1, \dots, \mathbf{v}_r \in V$	Elements $f_1, \dots, f_r \in M$
Scalars $c_1, \dots, c_r \in \mathbb{F}$	Coefficients $a_1, \dots, a_r \in R$
$c_1 \mathbf{v}_1 + \dots + c_r \mathbf{v}_r = \mathbf{0}$	$a_1 f_1 + \dots + a_r f_r = 0$
Linear dependence	Syzygy
$\ker(\mathbb{F}^r \rightarrow V)$	$\ker(R^r \rightarrow M)$
span of $\mathbf{v}_1, \dots, \mathbf{v}_r$	submodule generated by $f_1, \dots, f_r$

Feature	Linear Algebra	Commutative Algebra	Algebraic Geometry
Space	Vector space V over a field $\mathbb{F}$	Module/ideal over a ring R	Affine/projective space over a field
Objects	Vectors $(\mathbf{v}_1, \dots, \mathbf{v}_r)$	Generators $(f_1, \dots, f_r)$	Equations of a curve/surface
Coefficients	Scalars $c \in \mathbb{F}$	Ring elements $a \in R$	Regular functions
Relation	Linear relation	Syzygy	Syzygy
Set	Nullspace/kernel	Syzygy module	Syzygy sheaf/module
Geometric meaning	Linear subspaces through the origin	Constraints among generators	Hidden redundancies in defining equations
Complexity	Compute kernel	May need free resolutions	Linked to invariants such as Betti numbers, genus

1 Definition of Syzygy

Observation. The linear dependence of vectors $\mathbf{v}_1, \dots, \mathbf{v}_r \in V$ is expressed by

$$c_1 \mathbf{v}_1 + \dots + c_r \mathbf{v}_r = \mathbf{0}$$

for some scalars $c_1, \dots, c_r \in \mathbb{F}$.

In commutative algebra, one studies an analogous notion for elements of a ring, but now the coefficients are allowed to lie in the ring R itself.

Question. Let R be a ring. Suppose $f_1, \dots, f_r \in R$. Can we find elements $a_1, \dots, a_r \in R$ such that

$$a_1 f_1 + \dots + a_r f_r = 0 \in R?$$

If yes, then the tuple $(a_1, \dots, a_r)$ records a relation among the f_i . That relation is called a *syzygy*.

Syzygy

Definition 1. Let R be a commutative ring and let $f_1, \dots, f_r \in R$. A **syzygy** among $f_1, \dots, f_r$ is a tuple

$$(a_1, \dots, a_r) \in R^r \quad \text{such that} \quad a_1 f_1 + \dots + a_r f_r = 0.$$

Remark 1. This is richer than linear algebra because coefficients are no longer just constants; they may be polynomials or other ring elements.

Example 1. Let

$$f_1 = x, \quad f_2 = y \quad \text{where } f_1, f_2 \in \mathbb{C}[x, y].$$

A syzygy of f_1, f_2 is a pair $(a_1, a_2) \in \mathbb{C}[x, y]^2$ such that

$$a_1 x + a_2 y = 0.$$

The pair $(-y, x)$ is a syzygy of x and y , because

$$(-y)x + xy = -xy + xy = 0.$$

2 Syzygies as Kernels

Observation. Suppose $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_r \in V$ where V is a vector space over $\mathbb{F}$. Define

$$L := \{(c_1, c_2, \dots, c_r) \in \mathbb{F}^r : c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \dots + c_r\mathbf{v}_r = \mathbf{0}\} \subseteq \mathbb{F}^r.$$

Then L be a vector space:

(i) (Vector Addition) . Let $(c_1, c_2, \dots, c_r), (d_1, d_2, \dots, d_r) \in L$. Since

$$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \dots + c_r\mathbf{v}_r = \mathbf{0}, \quad d_1\mathbf{v}_1 + d_2\mathbf{v}_2 + \dots + d_r\mathbf{v}_r = \mathbf{0}$$

we have

$$(c_1 + d_1)\mathbf{v}_1 + (c_2 + d_2)\mathbf{v}_2 + \dots + (c_r + d_r)\mathbf{v}_r = \mathbf{0}.$$

(ii) (Scalar Multiplication). Let $\lambda \in \mathbb{F}$. Then

$$(\lambda c_1)\mathbf{v}_1 + (\lambda c_2)\mathbf{v}_2 + \dots + (\lambda c_r)\mathbf{v}_r = \lambda(c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \dots + c_r\mathbf{v}_r) = \lambda\mathbf{0} = \mathbf{0}.$$

Syzygy Module

Definition 2. The set of all syzygies among $f_1, \dots, f_r$ is called the *syzygy module* of $f_1, \dots, f_r$, denoted

$$\text{Syz}(f_1, \dots, f_r) = \{(a_1, \dots, a_r) \in R^r \mid a_1f_1 + \dots + a_rf_r = 0\}.$$

Remark 2. $\text{Syz}(f_1, \dots, f_r)$ is an R -submodule of R^r .

Proof. Suppose $f_1, \dots, f_r \in M$, where M is a R -module. Then $\text{Syz}(f_1, \dots, f_r)$ be a R -module:

- (i) Let $(a_1, \dots, a_r), (b_1, \dots, b_r) \in \text{Syz}(f_1, \dots, f_r)$. Since $\sum_{i=1}^r a_i f_i = 0$ and $\sum_{i=1}^r b_i f_i = 0$, then $\sum_{i=1}^r a_i f_i + \sum_{i=1}^r b_i f_i = \sum_{i=1}^r (a_i + b_i) f_i = 0$. So the sum is again a syzygy.
- (ii) Let $c \in R$. Then $\sum_{i=1}^r (ca_i) f_i = c \sum_{i=1}^r a_i f_i = c \dots 0 = 0$. So multiplying by an element of $\mathbb{R}$ gives another syzygy.

Hence $\text{Syz}(f_1, \dots, f_r)$ be a R -module. □

Example 2. Consider $R = \mathbb{C}[x, y]$. Then

$$\text{Syz}(x, y) = \mathbb{C}[x, y] \cdot (-y, x).$$

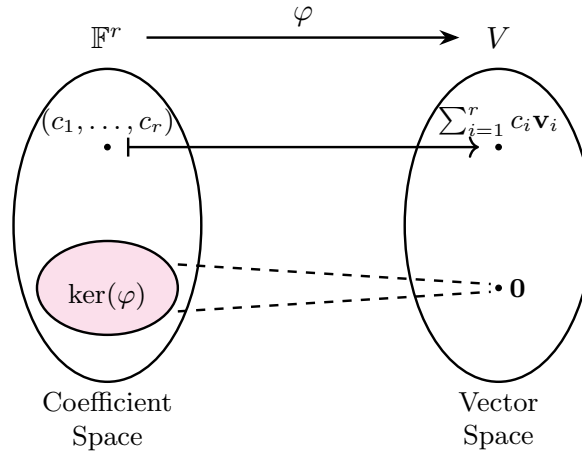
That is, every syzygy of x and y has the form $(-gy, gx)$ for some $g(x, y) \in \mathbb{C}[x, y]$.

Remark 3. There are infinitely many syzygies, but they are all generated by one basic syzygy.

Observation. Given vectors $\mathbf{v}_1, \dots, \mathbf{v}_r \in V$, define an $\mathbb{F}$ -linear map

$$\varphi : \mathbb{F}^r \rightarrow V \quad \text{by} \quad \varphi(c_1, \dots, c_r) = c_1 \mathbf{v}_1 + \dots + c_r \mathbf{v}_r.$$

Then the set of all linear relations among $\mathbf{v}_1, \dots, \mathbf{v}_r$ is $\ker(\varphi)$.



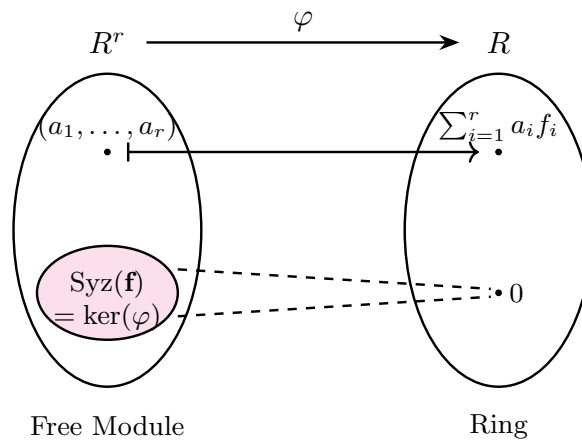
Observation. Given $f_1, \dots, f_r \in R$, define an R -linear map

$$\varphi : R^r \rightarrow R \quad \text{by} \quad \varphi(a_1, \dots, a_r) = a_1 f_1 + \dots + a_r f_r.$$

Then by definition,

$$\text{Syz}(f_1, \dots, f_r) = \ker(\varphi).$$

Thus a syzygy is a kernel element of the generator map.



3 Syzygies in Algebraic Geometry

In algebraic geometry, one can study a projective variety by studying the equations that define it inside projective space.

Example 3. Consider the map

$$\phi : \mathbb{P}^1 \rightarrow \mathbb{P}^3, \quad [s : t] \mapsto [s^3 : s^2t : st^2 : t^3].$$

Its image is called the *twisted cubic*. We write the homogeneous coordinates on $\mathbb{P}^3$ as

$$[x_0 : x_1 : x_2 : x_3],$$

Then points on the image satisfy

$$x_0 = s^3, \quad x_1 = s^2t, \quad x_2 = st^2, \quad x_3 = t^3.$$

We immediately get the quadratic relations

$$x_1^2 = x_0x_2, \quad x_1x_2 = x_0x_3, \quad x_2^2 = x_1x_3.$$

Thus the image lies on the three quadrics

$$f_1 = x_1^2 - x_0x_2, \quad f_2 =, \quad f_3 = x_2^2 - x_1x_3.$$

These equations define the twisted cubic. However, the equations themselves are not independent. Indeed,

$$\begin{aligned} x_2f_1 - x_1f_2 + x_0f_3 &= x_2(x_1^2 - x_0x_2) - x_1(x_1x_2 - x_0x_3) + x_0(x_2^2 - x_1x_3) \\ &= x_1x_1^2 - x_0x_2^2 - x_1^2x_2 + x_0x_1x_3 + x_0x_2^2 - x_0x_1x_3 \\ &= 0. \end{aligned}$$

So $(x_2, -x_1, x_0)$ is a syzygy among f_1, f_2, f_3 . Therefore,

- the equations f_1, f_2, f_3 describe the twisted cubic,
- the syzygy $(x_2, -x_1, x_0)$ describes how these equations fit together.

In algebraic geometry, the equations describe a variety, while syzygies describe relations among those equations.